

TM Forum Introductory Guide

AN L4 Wireless Energy Efficiency Optimization Solution Package

IG1526A

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Notice

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Table of Contents

Notice	2
Table of Contents	3
Executive Summary	4
Introduction.....	5
1. Scenario Description	6
1.1. AN L4 Wireless Energy efficiency optimization scenario categorization.....	6
2. Objectives	8
2.1. Objectives of AN L4 Wireless Energy efficiency optimization	8
2.2. Effectiveness Indicators of AN L4 Wireless Energy efficiency optimization	8
3. AN L4 Wireless Energy efficiency optimization Target Process and Key features.....	9
3.1. AN L4 Wireless Energy efficiency optimization evolution target ...	9
3.2. AN L4 Wireless Energy efficiency optimization key features.....	10
4. Architecture.....	12
4.1. Functional Architecture.....	12
5. Interaction Flow.....	15
5.1. AN L4 Wireless Energy efficiency optimization Interaction Flow.	15
5.2. AN L4 Wireless Energy efficiency optimization Interfaces	16
6. Conclusion	18
6.1. Interface standard requirement list.....	18
7. Terms & Abbreviations Used within this Document.....	19
7.1. Terminology	19
7.2. Abbreviations & Acronyms.....	19
8. References	20
9. Administrative Appendix	21
9.1. Document History	21
9.1.1. Version History.....	21
9.1.2. Release History.....	21
9.2. Acknowledgments.....	21

Executive Summary

The TM Forum Autonomous Networks (AN) Project has established a series of Introductory Guides and Guiding Books, including glossaries (such as IG1258), reference architectures (like IG1251), metric systems (IG1256), high-value scenario grading (IG1339), and assessment tools (IG1252 & GB1059). To assist users in quickly starting their AN Level 4 journey without needing extensive knowledge of the TM Forum AN Introductory Guides and Guiding Books and their background, the IG1500 series of documents provides instantiated descriptions that connect TM Forum Autonomous Networks standards, allowing users to intuitively understand the application and deployment of AN standards in real business scenarios.

This document defines a solution package for wireless energy efficiency optimization within the AN L4 framework, facilitating a swift initiation of AN L4 migration journey. It includes effectiveness indicators, evolution goals, reference architecture, and end-to-end workflows for wireless energy efficiency optimization scenario under the AN L4 structure.

Introduction

As the telecommunications industry evolves, Autonomous Network Level 4 (AN L4) is increasingly recognized as a consensus and goal for the development of operator networks. Many Communication Service Providers (CSPs) have already embarked on their journey to upgrade to AN L4. However, there is a noticeable lack of effective guidance on how to efficiently and rapidly establish AN L4 capabilities. In response to this need, the TM Forum's Autonomous Network Program (TMF ANP) has emerged, defining a series of Introductory Guides and Guiding Books to assist operators and vendors in their transition to AN L4.

This document presents a solution package for wireless energy efficiency optimization within the AN L4 framework. It includes effectiveness indicators, evolution goals, reference architecture, and end-to-end workflows for wireless energy efficiency optimization scenario under the AN L4 structure. This document focuses on energy efficiency optimization for AN L4 wireless network. Additionally, the integration modes of agent for wireless energy efficiency optimization is proposed.

1. Scenario Description

In modern mobile networks, the majority of energy consumption occurs in the field of Radio access networks (RAN). Among them, base station power consumption is one of the main reasons, accounting for about 80% of the total energy consumption. Operators are aiming at decreasing power consumption in 5G networks to lower their operational expense with energy saving management solutions. Traditional energy saving methods are mainly used by wireless network optimization engineers to save energy in the way of "manual judgment + periodic adjust energy saving actions (e.g. shutdown)", which can only achieve a certain degree of energy saving for a simple deployment scenario. However, the various combinations of energy saving actions can lead to conflicts especially for the complicated deployment scenarios (e.g. multiple RAT multiple frequencies' scenario). For example, different energy saving actions may be contradictory, or the energy saving actions may conflict with other activities (e.g. network optimization actions). Moreover, it is not straightforward to evaluate the influence on service experience (e.g. UL/DL RAN UE throughput, latency) of energy saving actions beforehand, which makes it difficult to balance the energy saving effect and service experience, for example the energy saving actions may deteriorate the service experience.

The intent driven approach for energy saving can enable the system to analyze and select the optimal balance between the energy saving effect and service experience by utilizing some intelligence mechanisms to achieve closed-loop automation for radio network energy saving. In intent driven approach, operator expresses intent expectation for RAN energy saving in the specified area (e.g. geographical area) to the system, which may include the RAN energy saving target (e.g. the maximum value of target RAN energy consumption, reduction ratio of energy consumption) and service experience (e.g. RAN UE throughput, latency), as well as the frequencies and RATs to be considered for energy saving, the system chooses an optimal value of RAN energy consumption to save energy as much as possible in the context to satisfy the service experience. The closed-loop automation for radio network energy saving involves key operation tasks, including setting energy saving targets, analysis of energy saving, formulating solutions, implementing solutions, and evaluating the effects of solution implementation.

1.1. AN L4 Wireless Energy efficiency optimization scenario categorization

According to the standard definitions in 3GPP TS 28.310, the AN L4 Wireless Energy efficiency optimization Sub-Scenario can be categorized into:

- Intra-RAT (Radio-Access-Technology) intra-frequency multi-Site energy saving: Represents energy saving across multiple sites using the same Radio Access Technology and operating on the same frequency band/carrier.

- Intra-RAT inter-frequency multi-Site energy saving: Represents energy saving across multiple sites using the same Radio Access Technology but operating on different frequencies band/carrier.
- Inter-RAT multi-Site energy saving: Represents energy saving across multiple sites using different Radio Access Technologies, including cases with both same and different frequencies band/carrier.

2. Objectives

2.1. Objectives of AN L4 Wireless Energy efficiency optimization

In the wireless energy efficiency optimization scenario, the objectives of AN L4 can be summarized as follows:

- 100% automation of wireless energy efficiency optimizations, fully automated translation of energy saving intents to formulate the scenario specific energy saving policies, analysis of energy saving, formulating solutions, implementing solutions, and evaluating the effects of solution implementation to achieve the intent driven closed loop.
- Hour-level dynamic orchestration and automatic execution of scenario policies.
- Optimal energy efficiency and service experience.

2.2. Effectiveness Indicators of AN L4 Wireless Energy efficiency optimization

In the wireless energy efficiency optimization scenario, to measure the effectiveness of AN L4, this article recommends following the indicators defined in IG1256 for a quantitative assessment of AN L4 performance. The relevant indicator definitions are as follows:

KCI and L4 Reference Values	Wireless energy efficiency optimization automation ratio: 100%
KEI and L4 Reference Values	Average Site level device Power-saving reduction ratio: XX% Average Site level Energy Reduction per GB: XX%
KBI and L4 Reference Values	Reduced O&M Costs: \$XX

3. AN L4 Wireless Energy efficiency optimization Target Process and Key features

3.1. AN L4 Wireless Energy efficiency optimization evolution target

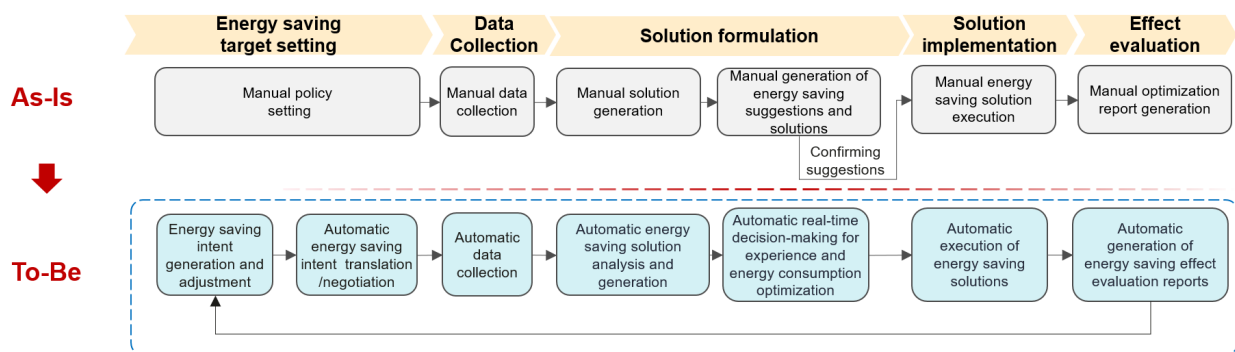


Figure 1. AN L4 Wireless Energy efficiency optimization evolution target

Following are the key transformation point:

- ① Energy saving policy formulation by experts → Automatic generation of energy saving policies based on intent translation
- ② Day-level energy saving → Dynamic energy saving, achieving real-time monitoring of traffic peak and off-peak hours and hour-level dynamic energy saving
- ③ Optimal energy consumption → Optimal experience and energy consumption, achieving online service experience assurance
- ④ Manual report generation → Automatic evaluation and presentation of energy savings effects

The intelligent capabilities of the wireless Energy efficiency optimization are provided in the following I-AADE Service Capabilities:

- **Energy saving target setting (Intent):** The System intelligently selects candidate energy-saving objects (e.g. NE/CELL) and generates the RAN energy saving policies based on the RAN energy saving intent. In addition, the system intelligently explores the energy saving gains and performance impacts to support continuous adjustment of intent targets to achieve optimal energy saving effect and service experience.

- **Effect evaluation (Intent):** The system intelligently evaluates and generates an intent fulfillment report, achieving automatic and accurate evaluation of energy saving effects.
- **Energy saving analysis (Awareness and Analysis):** The System intelligently predicts traffic load & network performance that can be used for energy saving solution and generates and adjust energy saving solutions to adapt to RAN Traffic & network Performance changes.
- **Solution formulation (Decision-making):** The System intelligently performs real-time decision-making (support the period of hours) for experience and energy consumption optimization suggestions as well as energy saving solution generation
- **Solution implementation (Execution):** The system automatically executes the energy saving solutions.

3.2. AN L4 Wireless Energy efficiency optimization key features

AN L4 Wireless Energy efficiency optimization defines key characteristics from five dimensions: energy saving target setting, energy saving analysis, solution formulation, solution implementation, and effect evaluation. Following are the key features to support intelligent capability by using any telecom models or advanced statistical calculations in above five dimensions.

- **Intent driven adaptive energy saving:** Based on the RAN energy saving intent, dynamically orchestrate multidimensional (i.e. Time domain, Spatial domain, Frequency domain and Power domain) energy saving features and generate the scenario specific energy saving policies to achieve energy saving even in peak hours for longer energy saving period. For example, the telecom model can be used for faster policy generation and decision-making to implement in-depth energy saving during busy hours. The multidimensional energy saving shutdown features includes symbol dormancy in time domain, channel shutdown in spatial domain, carrier in frequency domain and power coordination in power domain.
- **Digital twin-based multi-objective simulation and optimization:** Online simulation optimization iteration based on the digital twin environment to support real time coverage, traffic, service experience and energy consumption prediction and hour-level simulation optimization to ensure optimal service experience and Energy Consumption gain. Through analyzing the traffic and energy consumption, the energy consumption can be accurately evaluated with different shutdown scenarios.
- **Multi-frequency, multi-site, and multi-RAT coordinated energy saving.** Online simulation and dynamic orchestration of energy saving features across multiple sites using different radio access technologies and/or frequencies to achieve the optimal energy efficiency and service

experience for a RAN domain. For example, the telecom model can be used for inter sites co-coverage recognition and inter-RAT user migration to enhance NR carrier shutdown ratio.

- **Closed loop actuation:** automatically implement energy savings actions.

4. Architecture

This section describes the functional architecture of AN L4 Wireless Energy efficiency optimization, which follows a layered decoupling logic as defined in IG1251C. The integration modes of Copilot and Agent refer to the definitions provided in IG1251C.

4.1. Functional Architecture

Based on the preceding work division principles of each layer and the list of agents/copilot, and the general architecture of the AN L4 architecture (IG1251C), the architecture of the solution for wireless energy efficiency optimization scenario is as follows:

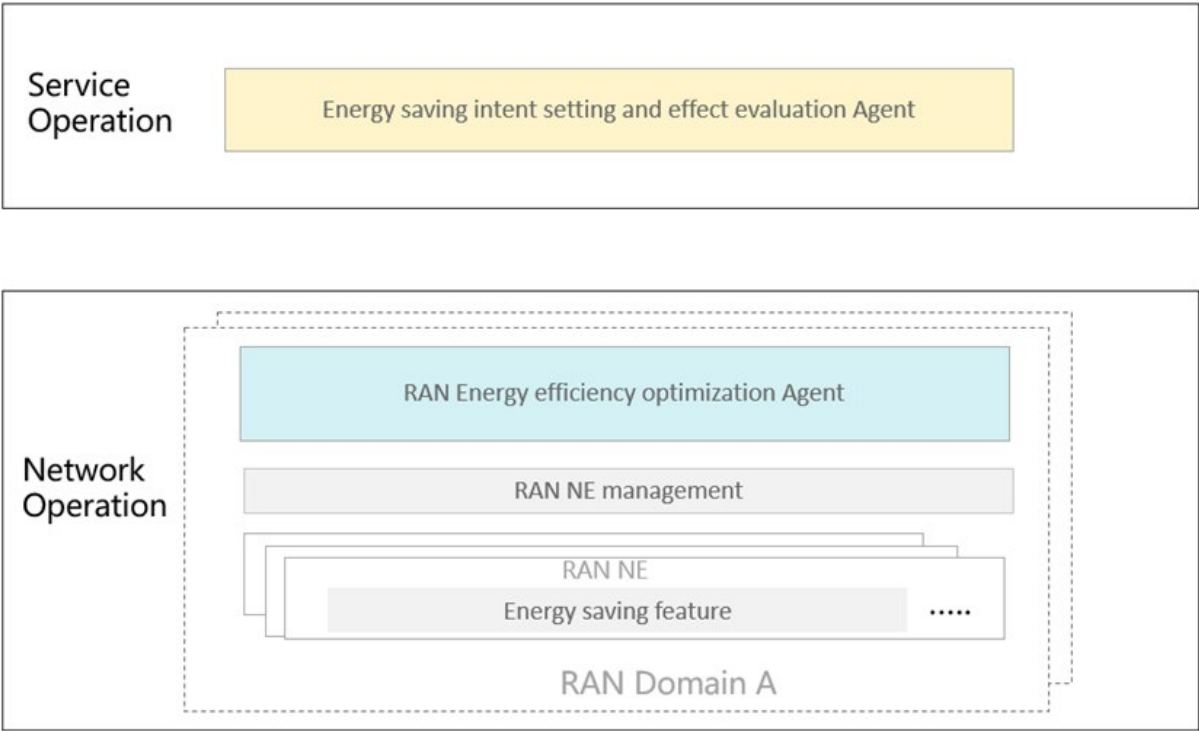


Figure 2. Architecture of the Wireless Energy efficiency optimization

In this architecture:

- Energy Saving Intent setting and Effect evaluation Agent in Service Operation layer, is a system focused on RAN energy saving intent generation, RAN energy saving effect evaluation and adjustment for different RAN Domains:
 - RAN energy saving intent generation, delivery and adjustment, analyze the business level energy saving requirements and generate RAN energy saving intents (e.g. RAN consumption targets, RAN UE throughput target and corresponding contexts) for different RAN Energy efficiency optimization Agents in different RAN domains. The RAN UE throughput target for different frequencies and RATs) in the same area maybe different. Before generating and delivering the RAN energy saving intent, it also can request the RAN Energy efficiency optimization Agent to explore the target values that are realistic and best aligned with RAN domain's capability. It also needs to dynamically adjust the intent targets and contexts (e.g. time period, RAT, frequency, and scenes) based on the RAN energy saving effect evaluation result and other information derived from the intent fulfillment report.
 - RAN energy saving effect evaluation, analyze the RAN energy saving effect for different RAN domains based on intent fulfillment report and KPIs from different Energy Saving Intent setting and Effect evaluation Agents.
- The RAN Energy efficiency optimization Agent in Network Operator Layer is an hour-level self-loop system focused on Wireless Energy efficiency optimization in a single RAN Domain, including following functionalities:
 - Energy saving intent exploration in pre-evaluation phase, explores the energy saving gains (e.g. Energy consumption reduction) and performance impacts (e.g. RAN UE throughput) to generate the intent exploration result. The intent exploration result can be used by operator to continuously adjust intent targets to achieve optimal energy saving effect and service experience (e.g. the best energy consumption target and RAN UE throughput target). The intent exploration will be performed based on RAN Domain Online [Simulation Capabilities](#), so the real network (including RAN NEs) will not be changed during the intent pre-evaluation phase.
 - Intent translation, selects candidate energy-saving objects (e.g. NE/CELL) of same or different RATs and Frequencies, dynamically orchestrate multidimensional energy saving features and generate the scenario specific energy saving policies based on the RAN energy saving intent.

- Automatic data collection **with data quality checks** and solution analysis, collects all RAN energy saving related data, predicts traffic load & performance that can be used for energy saving solution in real time and-generates and adjust the energy saving solutions to adapt to RAN Traffic & Performance changes.
- Automatic real-time decision-making for experience and energy consumption optimization suggestions. It intelligently evaluates energy saving solution when the recommended solutions are generated, and selects the best solution.
- Automatic execution of energy saving solutions, including configuring energy saving actions or policies for different RAN NEs.
- Automatic generation of energy saving effect evaluation and presentation, accurately evaluates and generates an intent fulfillment report including targets fulfillment status, corresponding achieved values (e.g. such as Energy consumption, RAN UE throughput). RAN NE configuration modification information can be provided in the intent fulfillment presentation as additional fulfillment information.
- RAN NEs provides service experience data to RAN Energy efficiency optimization Agent and execute energy saving actions or policies configured by RAN Energy efficiency optimization Agent.

5. Interaction Flow

Based on the above functional architecture and the integration model of Agent/Copilot, this chapter focuses on how to achieve end-to-end processing of the Wireless Energy efficiency optimization through the interaction and cooperation of agents at the Network operation layer and the Service operation layer.

5.1. AN L4 Wireless Energy efficiency optimization Interaction Flow

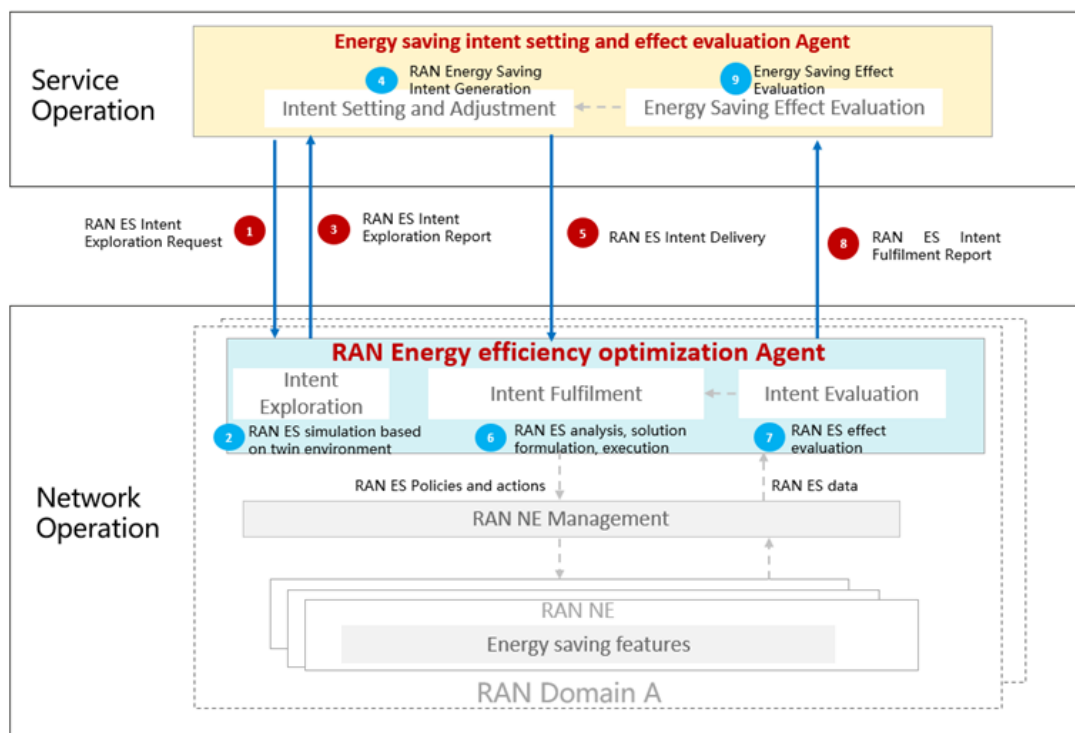


Figure 3. AN L4 Wireless Energy efficiency optimization Interaction Flow based on IAADE

For AN L4 Wireless Energy efficiency optimization, the RAN Energy efficiency optimization Agent operates in an autonomous closed-loop mode and interacts with the Energy Saving Intent setting and Effect evaluation Agent for cross RAN domains coordination, which includes two phases:

RAN Energy Saving intent pre-evaluation phase (Step 1- Step 3):

- Step 1 represents the request for RAN ES Intent Exploration. The request is triggered to explore the best values for RAN ES Intent targets (including RAN Energy consumption target and RAN UE throughput).
- Step 2 represents the exploration process to evaluate and determine the best possible values for the specified RAN ES Intent targets, considering current resource situation and capabilities of the system.
- Step3 represents the response to obtain the RAN ES exploration report, including the best possible values for the targets which are required in the Step1.

RAN Energy Saving intent fulfillment and evaluation phase (Step 4- Step 9):

- Step 4 represents RAN Energy Saving Intent generation process. In this process the RAN Energy Saving intent content is generated based on RAN ES exploration report and operator's business requirements. Different RAN UE throughput targets and RAN Energy Saving targets can be assigned for different frequencies or RATs (e.g. 4G, 5G) in different time periods.
- Step 5 represents the request to deliver the RAN ES intent.
- Step 6 represents a set of processes for RAN ES intent fulfillment, including RAN ES data collection, RAN ES analysis, solution formulation and execution (including perform the configuration activities for RAN ES policies and action for corresponding RAN NEs).
- Step 7 represents RAN ES effect evaluation and intent fulfillment report generation. The RAN ES intent fulfillment report may include the intent fulfillment status, achieved value and recommended values for corresponding intent targets and RAN NE configuration modification information.
- Step 8 represents the process to obtain the RAN ES intent fulfillment report.
- Step 9 represents the process to evaluate the Energy Saving effect evaluation for different RAN domains. Based on the Energy Saving effect evaluation result, the RAN ES intent may be adjusted.

5.2. AN L4 Wireless Energy efficiency optimization Interfaces

In the AN L4 Wireless Energy efficiency optimization, the following interface list is involved:

Interface No.	Interface name	Function Description of Interface Openness on the Network Side	Reference interface standard document	Related step no. in the process flow
Network-RAN ES Agent-API	RAN ES intent exploration interface	1. RAN ES intent exploration Request 2. RAN ES intent exploration result report	See the Intent exploration in Table 7.1-3 and RadioNetworkExpectation in clause 6.2.2 in 3GPP TS 28.312 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3554	Step 1 and Step 3
	RAN ES intent LCM interface	1. RAN ES intent creation 2. RAN ES intent modification 3. RAN ES intent deletion 4. RAN ES intent activation 5. RAN ES intent deactivation	See the intent lifecycle management in Table 7.1-1 and RadioNetworkExpectation in clause 6.2.2 in 3GPP TS 28.312 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3554	Step 5
	RAN ES intent report	1. Query an intent report 2. Subscribe an intent report 3. Unsubscribe an intent report 4. Notify an intent report	See the intent report management in Table 7.1-2 and IntentReport in clause 6.2.1.2.2 in 3GPP TS 28.312 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3554	Step8

6. Conclusion

6.1. Interface standard requirement list

Interface standard requirements in the AN L4 Wireless Energy efficiency optimization scenario, ref to 5.2.

7. Terms & Abbreviations Used within this Document

7.1. Terminology

Term	Definition	Source

7.2. Abbreviations & Acronyms

Abbreviation/Acronym	Abbreviation/Acronym Spelled Out	Definition	Source



8. References

Reference	Description	Source	Brief Use Summary

9. Administrative Appendix

9.1. Document History

9.1.1. Version History

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